

CHANG-KI BAEK

其他

基本信息

姓名: CHANG-KI BAEK

所属院校: 浦项理工大学

地区: South Korea

学科领域

微电子科学与工程

集成电路设计与集成系统

电子科学与技术

研究领域

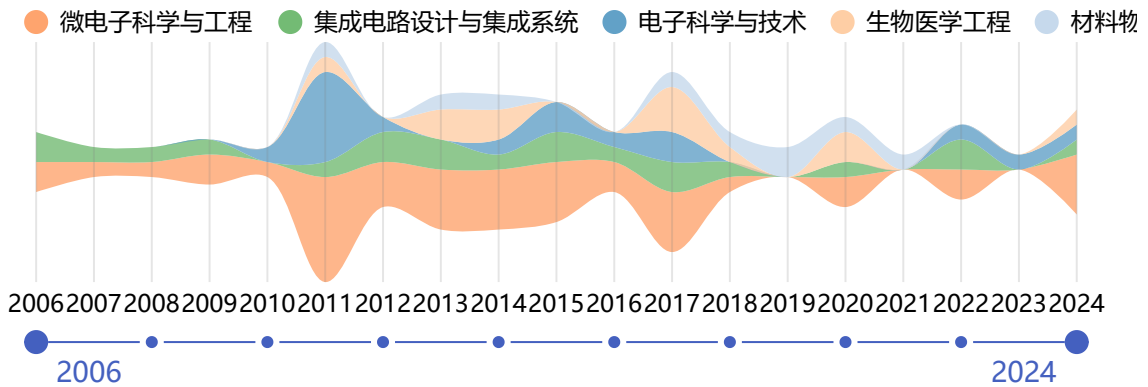
纳米电子与先进工艺

微电子器件与制造

个人简介

CHANG - KI BAEK is a professor at the Pohang University of Science and Technology (POSTECH). His research areas include Electronics, Nanotechnology, Energy, and Sensor. He has an overall h - index of 28 and i10 - index of 60, with 2554 bibliographic records. After 2020, his h - index is 20, i10 - index is 36, and there are 1511 bibliographic records. He has co - authored papers with many researchers from institutions like NASA Ames Research Center, TU Dresden, etc. Some of his well - cited papers are "A proposal on an optimized device structure with experimental studies on recent devices for the DRAM cell transistor" (278 citations in 2007) and "Electrical characteristics of 20 - nm junctionless Si nanowire transistors" (179 citations in 2012).

近年发表的论文



Schottky barrier memory based on heterojunction bandgap engineering for high-density and low-power retention

DISCOVER NANO

Kim, Hyangwoo; Kim, Yijoon; Oh, Kyoungghan; Park, Ju Hong; Baek, Chang-Ki

被引用次数: 0

2024

Enhancing Radio Frequency Performance of Graphene Field-Effect Transistors through Machine-Learning-Based Physical Prediction and Optimization

ACS APPLIED ELECTRONIC MATERIALS

Seo, Gyeong Min; Baek, Chang-Ki; Kong, Byoung Don

被引用次数: 0

2024

Neuromorphic dendritic network computation with silent synapses for visual motion perception

NATURE ELECTRONICS

Baek, Eunhye; Song, Sen; Baek, Chang-Ki; Rong, Zhao; Shi, Luping; Cannistraci, Carlo Vittorio

被引用次数: 3

2024

New tunneling source follower with low 1/f noise and high voltage gain

SCIENTIFIC REPORTS

Kim, Ki Yeong; Kim, Hyangwoo; Choi, Minkeun; Oh, Kyoungghan; Yoo, Hyeongseok; Kim, Yijoon; Sol, Yoon; Park, Sungbong; Lee, Jae-Kyu; Baek, Chang-Ki

被引用次数: 0

2024

Highly biomimetic spiking neuron using SiGe heterojunction bipolar transistors for energy-efficient neuromorphic systems

SCIENTIFIC REPORTS

Kim, Yijoon; Kim, Hyangwoo; Oh, Kyoungghan; Park, Ju Hong; Baek, Chang-Ki

被引用次数: 1

2024

High-voltage FinFET with floating poly and high-k material for enhanced intrinsic gain and safe operating area

SCIENTIFIC REPORTS

Oh, Kyoungghan; Kim, Hyangwoo; Kim, Yijoon; Yoo, Hyeongseok; Choi, Minkeun; Maeng, Wooyeol; Kim, Suta; Lee, Hyung-Jin; Baek, Chang-Ki

被引用次数: 0

2024

Low-energy and tunable LIF neuron using SiGe bandgap-engineered resistive switching transistor

DISCOVER NANO

Kim, Yijoon; Kim, Hyangwoo; Oh, Kyoungghan; Park, Ju Hong; Kong, Byoung Don; Baek, Chang-Ki

被引用次数: 0

2024

Hydrogen Fluoride Gas Sensor by Silicon Nanosheet Field-Effect Transistor

IEEE SENSORS JOURNAL

Kwak, Hyeon-Tak; Kim, Hyangwoo; Yoo, Hyeongseok; Choi, Minkeun; Oh, Kyoungghan; Kim, Yijoon; Kong, Byoung Don; Baek, Chang-Ki

被引用次数: 2

2023

Enhanced thermoelectric figure of merit in highly-doped silicon nanowires via a corrugated surface modulation

NANO ENERGY

Yoo, Hyeongseok; Kim, Hyangwoo; Kwak, Hyeon-Tak; Choi, Minkeun; Oh, Kyoungghan; Kim, Yijoon; Kim, Ki Yeong; Lee, Seungho; Kong, Byoung Don; Park, Ju Hong; Baek, Chang-Ki

被引用次数: 1

2023

Fluoride ion and hydrofluoric acid detection via silicon nanosheet field-effect transistor sensor

SENSORS AND ACTUATORS B-CHEMICAL

Kwak, Hyeon-Tak; Kim, Hyangwoo; Yoo, Hyeongseok; Choi, Minkeun; Kong, Byoung Don; Baek, Chang-Ki

被引用次数: 1

2023

Enhanced near-infrared photodetection via whispering gallery modes in the wave-shaped sidewall silicon nanopillar arrays

OPTICS EXPRESS

Choi, Minkeun; Kwak, Hyeon-Tak; Kim, Hyangwoo; Yoo, Hyeongseok; Park, Ju Hong; Baek, Chang-Ki

被引用次数: 0

2023

A drain extended FinFET with enhanced DC/RF performance for high-voltage RF applications

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Oh, Kyoungghan; Kim, Hyangwoo; Park, Kangwook; Lee, Hyung-jin; Kong, Byoung Don; Baek, Chang-Ki

被引用次数: 1

2022

mm-band surface acoustic wave devices utilizing two-dimensional boron nitride

SCIENTIFIC REPORTS

Yoon, Seok Hyun; Baek, Chang-Ki; Kong, Byoung Don

被引用次数: 0

2022

Highly Reliable Memory Operation of High-Density Three-Terminal Thyristor Random Access Memory

NANOSCALE RESEARCH LETTERS

Kim, Hyangwoo; Cho, Hyeonsu; Kwak, Hyeon-Tak; Seo, Myunghae; Lee, Seungho; Kong, Byoung Don; Baek, Chang-Ki

被引用次数: 1

2022

Enhanced Thermoelectric Properties of Cobalt Silicide-Silicon Heterostructured Nanowires

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Lee, Seungho; Cho, Hyeonsu; Yoon, Sol; Seo, Myunghae; Yoo, Hyeongseok; Kong, Byoung Don; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 4

2021

Two-Dimensional Boronate Ester Covalent Organic Framework Thin Films with Large Single Crystalline Domains for a Neuromorphic Memory Device

ANGEWANDTE CHEMIE-INTERNATIONAL EDITION

Park, SangWook; Liao, Zhongquan; Ibarlucea, Bergoi; Qi, Haoyuan; Lin, Hung-Hsuan; Becker, Daniel; Melidonie, Jason; Zhang, Tao; Sahabudeen, Hafeesudeen; Baraban, Larysa; Baek, Chang-Ki; Zheng, Zhikun; Zschech, Ehrenfried; Fery, Andreas; Heine, Thomas; Kaiser, Ute; Cuniberti, Gianaurelio; Dong, Renhao; Feng, Xinliang

被引用次数: 99

2020

Nanosensor-Based Real-Time Monitoring of Stress Biomarkers in Human Saliva Using a Portable Measurement System

ACS SENSORS

Klinghammer, Stephanie; Voitsekhivska, Tetiana; Licciardello, Nadia; Kim, Kihyun; Baek, Chang-Ki; Cho, Hyeonsu; Wolter, Klaus-Jurgen; Kirschbaum, Clemens; Baraban, Larysa; Cuniberti, Gianaurelio

被引用次数: 32

2020

Buffer solution optimization for accurate fluoride ion detection in tap water

JOURNAL OF ELECTROANALYTICAL CHEMISTRY

Cho, Hyeonsu; Kim, Kihyun; Kwak, Hyeon-Tak; Lee, Seungho; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 9

2020

Disorder Originated Unusual Mobility in Crystalline InGaZnO₄

IEEE ELECTRON DEVICE LETTERS

Jegal, Yoon; Kulahlioglu, Adem H.; Baek, Chang-Ki; Kong, Byoung Don

被引用次数: 4

2020

High-Voltage Drain-Extended FinFET With a High-k Dielectric Field Plate

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Hyangwoo; Cho, Hyeonsu; Kong, Byoung Don; Park, Kangwook; Kim, Ilryong; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 8

2020

Solar Cell Using Hourglass-Shaped Silicon Nanowires for Increased Light-Trapping Path (vol 10, pg 475, 2020)

IEEE JOURNAL OF PHOTOVOLTAICS

Seo, Myunghae; Yoon, Sol; Cho, Hyeonsu; Lee, Seungho; Kim, Kihyun; Kong, Byoung Don; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 0

2020

Solar Cell Using Hourglass-Shaped Silicon Nanowires for Increased Light-Trapping Path

IEEE JOURNAL OF PHOTOVOLTAICS

Seo, Myunghae; Yoon, Sol; Cho, Hyeonsu; Lee, Seunggho; Kim, Kihyun; Kong, Byoung Don; Meyyappan, Meyya; Baek, Chang-Ki

被引用次数: 16

2020

Intrinsic plasticity of silicon nanowire neurotransistors for dynamic memory and learning functions

NATURE ELECTRONICS

Baek, Eunhye; Das, Nikhil Ranjan; Cannistraci, Carlo Vittorio; Rim, Taiuk; Bermudez, Gilbert Santiago Canon; Nych, Khrystyna; Cho, Hyeonsu; Kim, Kihyun; Baek, Chang-Ki; Makarov, Denys; Tetzlaff, Ronald; Chua, Leon; Baraban, Larysa; Cuniberti, Gianaurelio

被引用次数: 35

2020

Thermal conductivity reduction by scallop shaped surface modulation in silicon nanowires

APPLIED PHYSICS LETTERS

Lee, Seunggho; Yoo, Hyeongseok; Won, Woo-Yeon; Cho, Hyeonsu; Seo, Myunghae; Kong, Byoung Don; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 8

2020

Usefulness of the Assessment of CT-Based Donor Kidney Function Compared with Radionuclide Scintigraphy.

AMERICAN JOURNAL OF TRANSPLANTATION

Park, K.; Yoon, J.; Kim, H.; Baek, C.; Park, S.; Kim, H.

被引用次数: 0

2019

Design Guidelines for High Sensitivity ZnO Nanowire Gas Sensors With Schottky Contact

IEEE SENSORS JOURNAL

Kim, Kihyun; Kwak, Hyeon-Tak; Cho, Hyeonsu; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 9

2019

Whispering gallery modes enhance the near-infrared photoresponse of hourglass-shaped silicon nanowire photodiodes

NATURE ELECTRONICS

Kim, Kihyun; Yoon, Sol; Seo, Myunghae; Lee, Seunggho; Cho, Hyeonsu; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 39

2019

Vertical Silicon Nanowire Thermoelectric Modules with Enhanced Thermoelectric Properties

NANO LETTERS

Lee, Seunggho; Kim, Kihyun; Kang, Deok-Hong; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 46

2019

LaF3 electrolyte-insulator-semiconductor sensor for detecting fluoride ions

SENSORS AND ACTUATORS B-CHEMICAL

Cho, Hyeonsu; Kim, Kihyun; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 29

2019

Weakly Tapered Silicon Nanopillar Resonators with Spatially Well Separated Whispering Gallery Modes for Si-Based Lasers

ACS APPLIED NANO MATERIALS

Seo, Myunghae; Kim, Kihyun; Cho, Hyeonsu; Yoon, Sol; Kong, Byoung Don; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 6

2019

An array of metal oxides nanoscale hetero p-n junctions toward designable and highly-selective gas sensors

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Hyunah; Yoon, Jun-Sik; Lee, Yuna; Kim, Dong Yeong; Baek, Chang-Ki; Kim, Jong Kyu

被引用次数: 39

2018

Ultrasensitive detection of Ebola matrix protein in a memristor mode

NANO RESEARCH

Ibarlucea, Bergoi; Akbar, Teuku Fawzul; Kim, Kihyun; Rim, Taiuk; Baek, Chang-Ki; Ascoli, Alon; Tetzlaff, Ronald; Baraban, Larysa; Cuniberti, Gianaurelio

被引用次数: 39

2018

Bandgap Engineering and Strain Effects of Core-Shell Tunneling Field-Effect Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yoon, Jun-Sik; Kim, Kihyun; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 12

2018

Negative Photoconductance in Heavily Doped Si Nanowire Field-Effect Transistors

NANO LETTERS

Baek, Eunhye; Rim, Taiuk; Schutt, Julian; Baek, Chang-ki; Kim, Kihyun; Baraban, Larysa; Cuniberti, Gianaurelio

被引用次数: 77

2017

Core-shell homojunction silicon vertical nanowire tunneling field-effect transistors

SCIENTIFIC REPORTS

Yoon, Jun-Sik; Kim, Kihyun; Baek, Chang-Ki

被引用次数: 13

2017

Polysilicon near-infrared photodetector with performance comparable to crystalline silicon devices

OPTICS EXPRESS

Yoon, Sol; Kim, Kihyun; Cho, Hyeonsu; Yoon, Jun-Sik; Lee, Myoung Jin; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 7

2017

Nanowire sensors monitor bacterial growth kinetics and response to antibiotics

LAB ON A CHIP

Ibarlucea, B.; Rim, T.; Baek, C. K.; de Visser, J. A. G. M.; Baraban, L.; Cuniberti, G.

Performance and Variations Induced by Single Interface Trap of Nanowire FETs at 7-nm Node

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yoon, Jun-Sik; Kim, Kihyun; Rim, Taiuk; Baek, Chang-Ki

被引用次数: 22

2017

Optical Characteristics of Silicon-Based Asymmetric Vertical Nanowire Photodetectors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yoon, Jun-Sik; Kim, Kihyun; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 12

2017

Electrical Characteristics of Doped Silicon Nanowire Channel Field-Effect Transistor Biosensors

IEEE SENSORS JOURNAL

Rim, Taiuk; Kim, Kihyun; Cho, Hyeonsu; Jeong, Wooju; Yoon, Jun-Sik; Kim, Yumi; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 13

2017

Optimization of Signal to Noise Ratio in Silicon Nanowire ISFET Sensors

IEEE SENSORS JOURNAL

Cho, Hyeonsu; Kim, Kihyun; Yoon, Jun-Sik; Rim, Taiuk; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 13

2017

Process-Induced Variations of 10-nm Node Bulk nFinFETs Considering Middle-of-Line Parasitics

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yoon, Jun-Sik; Baek, Chang-Ki; Baek, Rock-Hyun

被引用次数: 23

2016

Effects of single grain boundary and random interface traps on electrical variations of sub-30 nm polysilicon nanowire structures

MICROELECTRONIC ENGINEERING

Oh, Hyeongwan; Kim, Jungsik; Lee, Junyoung; Rim, Taiuk; Baek, Chang-Ki; Lee, Jeong-Soo

被引用次数: 13

2016

Variability study of Si nanowire FETs with different junction gradients

AIP ADVANCES

Yoon, Jun-Sik; Kim, Kihyun; Rim, Taiuk; Baek, Chang-Ki

被引用次数: 10

2016

Three-dimensional simulation of threshold voltage variations due to an oblique single grain boundary in sub-40nm polysilicon nanowire FETs

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Jungsik; Oh, Hyeongwan; Lee, Junyoung; Baek, Chang-Ki; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 5

2015

Statistical variability study of random dopant fluctuation on gate-all-around inversion-mode silicon nanowire field-effect transistors

APPLIED PHYSICS LETTERS

Yoon, Jun-Sik; Rim, Taiuk; Kim, Jungsik; Kim, Kihyun; Baek, Chang-Ki; Jeong, Yoon-Ha

被引用次数: 30

2015

High efficiency silicon solar cell based on asymmetric nanowire

SCIENTIFIC REPORTS

Ko, Myung-Dong; Rim, Taiuk; Kim, Kihyun; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 56

2015

Junction Design Strategy for Si Bulk FinFETs for System-on-Chip Applications Down to the 7-nm Node

IEEE ELECTRON DEVICE LETTERS

Yoon, Jun-Sik; Jeong, Eui-Young; Baek, Chang-Ki; Kim, Ye-Ram; Hong, Jae-Ho; Lee, Jeong-Soo; Baek, Rock-Hyun; Jeong, Yoon-Ha

被引用次数: 35

2015

Investigation of RC Parasitics Considering Middle-of-the-Line in Si-Bulk FinFETs for Sub-14-nm Node Logic Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jeong, Eui-Young; Yoon, Jun-Sik; Baek, Chang-Ki; Kim, Ye-Ram; Hong, Jae-Ho; Lee, Jeong-Soo; Baek, Rock-Hyun; Jeong, Yoon-Ha

被引用次数: 15

2015

Chemical Gated Field Effect Transistor by Hybrid Integration of One-Dimensional Silicon Nanowire and Two-Dimensional Tin Oxide Thin Film for Low Power Gas Sensor

ACS APPLIED MATERIALS & INTERFACES

Han, Jin-Woo; Rim, Taiuk; Baek, Chang-Ki; Meyyappan, M.

被引用次数: 50

2015

Universal relaxation characteristic of interface trap under FN and NBTI stress in pMOSFET device

ELECTRONICS LETTERS

Choi, Seongwook; Park, Sooyoung; Baek, Chang-Ki; Park, Young June

被引用次数: 1

2014

Vertical gate-all-around junctionless nanowire transistors with asymmetric diameters and underlap lengths

APPLIED PHYSICS LETTERS

Yoon, Jun-Sik; Rim, Taiuk; Kim, Jungsik; Meyyappan, Meyya; Baek, Chang-Ki; Jeong, Yoon-Ha

被引用次数: 32

2014

Silicon Nanowire Biologically Sensitive Field Effect Transistors: Electrical Characteristics and Applications

Threshold Voltage Variations Due to Oblique Single Grain Boundary in Sub-50-nm Polysilicon Channel

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Jungsik; Rim, Taiuk; Lee, Junyoung; Baek, Chang-Ki; Meyyappan, Meyya; Lee, Jeong-Soo

被引用次数: 12

2014

Optimized operation of silicon nanowire field effect transistor sensors

NANOTECHNOLOGY

Rim, Taiuk; Meyyappan, M.; Baek, Chang-Ki

被引用次数: 10

2014

Thermally Phase-Transformed In₂Se₃ Nanowires for Highly Sensitive Photodetectors

SMALL

Kang, Daegun; Rim, Taiuk; Baek, Chang-Ki; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 30

2014

Investigation of Low-Frequency Noise in p-type Nanowire FETs: Effect of Switched Biasing Condition and Embedded SiGe Layer

IEEE ELECTRON DEVICE LETTERS

Lee, Sang-Hyun; Kim, Ye-Ram; Hong, Jae-Ho; Jeong, Eui-Young; Yoon, Jun-Sik; Baek, Chang-Ki; Kim, Dong-Won; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 4

2014

Simple S/D Series Resistance Extraction Method Optimized for Nanowire FETs

IEEE ELECTRON DEVICE LETTERS

Kim, Ye-Ram; Lee, Sang-Hyun; Sohn, Chang-Woo; Choi, Do-Young; Sagong, Hyun-Chul; Kim, Sungho; Jeong, Eui-Young; Kim, Dong-Won; Hong, Hyeongsun; Baek, Chang-Ki; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 8

2013

Investigation of electromigration in In₂Se₃ nanowire for phase change memory devices

APPLIED PHYSICS LETTERS

Kang, Daegun; Rim, Taiuk; Baek, Chang-Ki; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 16

2013

Analytic Model of S/D Series Resistance in Trigate FinFETs With Polygonal Epitaxy

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sohn, Chang-Woo; Kang, Chang Yong; Ko, Myung-Dong; Choi, Do-Young; Sagong, Hyun Chul; Jeong, Eui-Young; Park, Chan-Hoon; Lee, Sang-Hyun; Kim, Ye-Ram; Baek, Chang-Ki; Lee, Jeong-Soo; Lee, Jack C.; Jeong, Yoon-Ha

被引用次数: 24

2013

Improved performance of In₂Se₃ nanowire phase-change memory with SiO₂ passivation

SOLID-STATE ELECTRONICS

Baek, Chang-Ki; Kang, Daegun; Kim, JeongSik; Jin, Bo; Rim, Taiuk; Park, Sooyoung; Meyyappan, M.; Jeong, Yoon-Ha; Lee, Jeong-Soo

被引用次数: 19

2013

Improved Electrical Characteristics of Honeycomb Nanowire ISFETs

IEEE ELECTRON DEVICE LETTERS

Rim, Taiuk; Kim, Kihyun; Kim, Sungho; Baek, Chang-Ki; Meyyappan, Meyya; Jeong, Yoon-Ha; Lee, Jeong-Soo

被引用次数: 23

2013

Investigation of the electrical stability of Si-nanowire biologically sensitive field-effect transistors with embedded Ag/AgCl pseudo reference electrode

RSC ADVANCES

Rim, Taiuk; Kim, Kihyun; Hong, Nanki; Ko, Wooree; Baek, Chang-Ki; Jeon, Sangmin; Deen, M. Jamal; Meyyappan, M.; Jeong, Yoon-Ha; Lee, Jeong-Soo

被引用次数: 24

2013

Study on a Scaling Length Model for Tapered Tri-Gate FinFET Based on 3-D Simulation and Analytical Analysis

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ko, Myung-Dong; Sohn, Chang-Woo; Baek, Chang-Ki; Jeong, Yoon-Ha

被引用次数: 36

2013

Electrical characteristics of 20-nm junctionless Si nanowire transistors

SOLID-STATE ELECTRONICS

Park, Chan-Hoon; Ko, Myung-Dong; Kim, Ki-Hyun; Baek, Rock-Hyun; Sohn, Chang-Woo; Baek, Chang Ki; Park, Sooyoung; Deen, M. J.; Jeong, Yoon-Ha; Lee, Jeong-Soo

被引用次数: 133

2012

Characterization of Channel-Diameter-Dependent Low-Frequency Noise in Silicon Nanowire Field-Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Lee, Sang-Hyun; Baek, Chang-Ki; Park, Sooyoung; Kim, Dong-Won; Sohn, Dong Kyun; Lee, Jeong-Soo; Kim, Dae M.; Jeong, Yoon-Ha

被引用次数: 28

2012

Device Design Guidelines for Nanoscale FinFETs in RF/Analog Applications

IEEE ELECTRON DEVICE LETTERS

Sohn, Chang-Woo; Kang, Chang Yong; Baek, Rock-Hyun; Choi, Do-Young; Sagong, Hyun Chul; Jeong, Eui-Young; Baek, Chang-Ki; Lee, Jeong-Soo; Lee, Jack C.; Jeong, Yoon-Ha

被引用次数: 13

2012

Optical and electrical characteristics of asymmetric nanowire solar cells

JOURNAL OF APPLIED PHYSICS

Ko, Myung-Dong; Baek, Chang-Ki; Rim, Taiuk; Park, Sooyoung; Jeong, Yoon-Ha

被引用次数: 12

2012

C-V Characteristics in Undoped Gate-All-Around Nanowire FET Array

IEEE ELECTRON DEVICE LETTERS

Baek, Rock-Hyun; Baek, Chang-Ki; Lee, Sang-Hyun; Suk, Sung Dae; Li, Ming; Yeoh, Yun Young; Yeo, Kyoung Hwan; Kim, Dong-Won; Lee, Jeong-Soo; Kim, Dae M.; Jeong, Yoon-Ha

被引用次数: 11

2011

New Investigation of Hot-Carrier Degradation on RF Small-Signal Parameter and Performance in High-k/Metal-Gate nMOSFETs

IEEE ELECTRON DEVICE LETTERS

Sagong, Hyun Chul; Kang, Chang Yong; Sohn, Chang-Woo; Choi, Do-Young; Jeong, Eui-Young; Baek, Chang-Ki; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 0

2011

Comprehensive Study of Quasi-Ballistic Transport in High- κ /Metal Gate nMOSFETs

IEEE ELECTRON DEVICE LETTERS

Sagong, Hyun Chul; Kang, Chang Yong; Sohn, Chang-Woo; Jeon, Kanghoon; Jeong, Eui-Young; Choi, Do-Young; Baek, Chang-Ki; Lee, Jeong-Soo; Lee, Jack C.; Jeong, Yoon-Ha

被引用次数: 2

2011

Silicon nanowire ion sensitive field effect transistor with integrated Ag/AgCl electrode: pH sensing and noise characteristics

ANALYST

Kim, Sungho; Rim, Taiuk; Kim, Kihyun; Lee, Unsang; Baek, Eunhye; Lee, Hojoon; Baek, Chang-Ki; Meyyappan, M.; Deen, M. Jamal; Lee, Jeong-Soo

被引用次数: 77

2011

Interfacial-Layer-Driven Dielectric Degradation and Breakdown of HfSiON/SiON Gate Dielectric nMOSFETs

IEEE ELECTRON DEVICE LETTERS

Choi, Do-Young; Lee, Kyong Taek; Baek, Chang-Ki; Sohn, Chang Woo; Sagong, Hyun Chul; Jung, Eui-Young; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 10

2011

Characterization and Modeling of 1/f Noise in Si-nanowire FETs: Effects of Cylindrical Geometry and Different Processing of Oxides

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Baek, Rock-Hyun; Baek, Chang-Ki; Choi, Hyun-Sik; Lee, Jeong-Soo; Yeoh, Yun Young; Yeo, Kyoung Hwan; Kim, Dong-Won; Kim, Kinam; Kim, Dae M.; Jeong, Yoon-Ha

被引用次数: 21

2011

A 3-D Statistical Simulation Study of Mobility Fluctuations in MOSFET Induced by Discrete Trapped Charges in SiO₂ Layer

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Park, Sooyoung; Baek, Chang-Ki; Park, Hong-Hyun; Choi, SeongWook; Park, Young June

被引用次数: 1

2011

An Analysis of the Field Dependence of Interface Trap Generation under Negative Bias Temperature Instability Stress using Wentzel-Kramers-Brillouin with Density Gradient Method

JAPANESE JOURNAL OF APPLIED PHYSICS

Choi, SeongWook; Baek, Chang-Ki; Park, Sooyoung; Park, Young June

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2011

Characteristics of the Series Resistance Extracted From Si Nanowire FETs Using the Y-Function Technique

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Baek, Rock-Hyun; Baek, Chang-Ki; Jung, Sung-Woo; Yeoh, Yun Young; Kim, Dong-Won; Lee, Jeong-Soo; Kim, Dae M.; Jeong, Yoon-Ha

被引用次数: 26

2010

Characterization of Near-Interface Oxide Trap Density in Nitrided Oxides for Nanoscale MOSFET Applications

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Son, Younghwan; Baek, Chang-Ki; Han, In-Shik; Joo, Han-Soo; Goo, Tae-Gyu; Yoo, Ooksang; Choi, Wonho; Ji, Hee-Hwan; Lee, Hi-Deok; Kim, Dae M.

被引用次数: 4

2009

A comparative study of the DRAM leakage mechanism for planar and recessed channel MOSFETs

SOLID-STATE ELECTRONICS

Lee, Myoung Jin; Baek, Chang-Ki; Park, Sooyoung; Chung, In-Young; Park, Young June

被引用次数: 5

2009

Three-dimensional simulation of dopant-fluctuation-induced threshold voltage dispersion in nonplanar MOS structures targeting flash EEPROM transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Bomsoo; Kwon, Wookhyun; Baek, Chang-Ki; Jin, Seonghoon; Song, Yunheub; Kim, Dae M.

被引用次数: 4

2008

A proposal on an optimized device structure with experimental studies on recent devices for the DRAM cell transistor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Myoung Jin; Jin, Seonghoon; Baek, Chang-Ki; Hong, Sung-Min; Park, Soo-Young; Park, Hong-Hyun; Lee, Sang-Don; Chung, Sung-Woong; Jeong, Jae-Goan; Hong, Sung-Joo; Park, Sung-Wook; Chung, In-Young; Park, Young June; Min, Hong Shick

被引用次数: 195

2007

Reliable extraction of cycling induced interface states implementing realistic P/E stresses in reference cell: Comparison with flash memory cell

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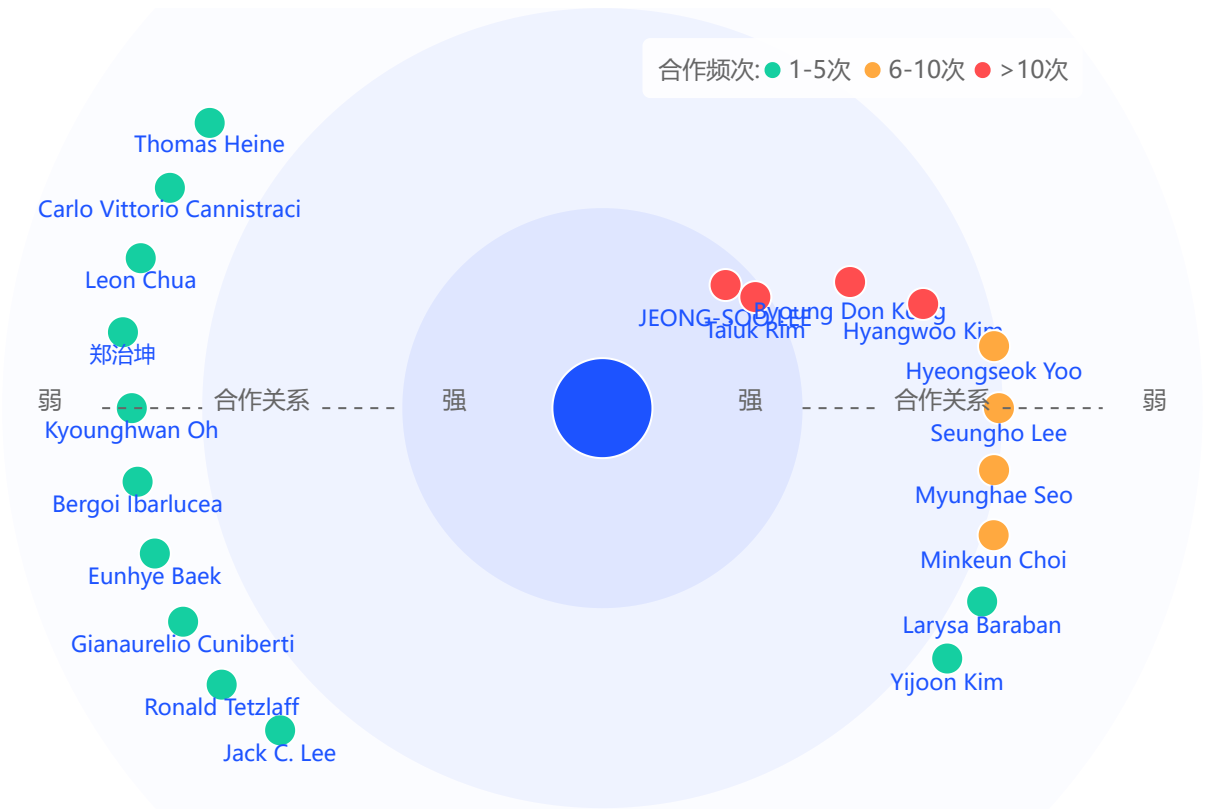
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